Machine Learning–Based Intrusion Detection Solution for Securing Vehicular Ad-hoc Networks (VANETs)

Group No. 3
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Problems

- VANET vulnerability to attacks such as malware, Sybil, DoS, and false data injection.

- Disruption to traffic safety applications, compromise driver privacy and even cause accidents.

- Existing security measures not enough; advanced Intrusion Detection Systems (IDSs) are needed.

Solution

Using SVM (Support Vector Machine) to find malicious packets in our VANET network.

- Finding out malicious nodes is difficult reasons being
  - Lack of central points with peer-to-peer communication
  - High mobility of nodes.
  - Wireless links with limited network infrastructure.
- Analogy - Imagine a group of cyclists (vehicles) riding around a city, each with walkie-talkies (wireless links) talking to each other directly (P2P). There's no central dispatcher. If one cyclist pretends to be someone else or sends wrong directions, others might follow — and there's no easy way to verify or stop it.
- Why SVM ? - SVM is best suited for datasets that are highly dimensional and is also robust to noise where overfitting is minimized.

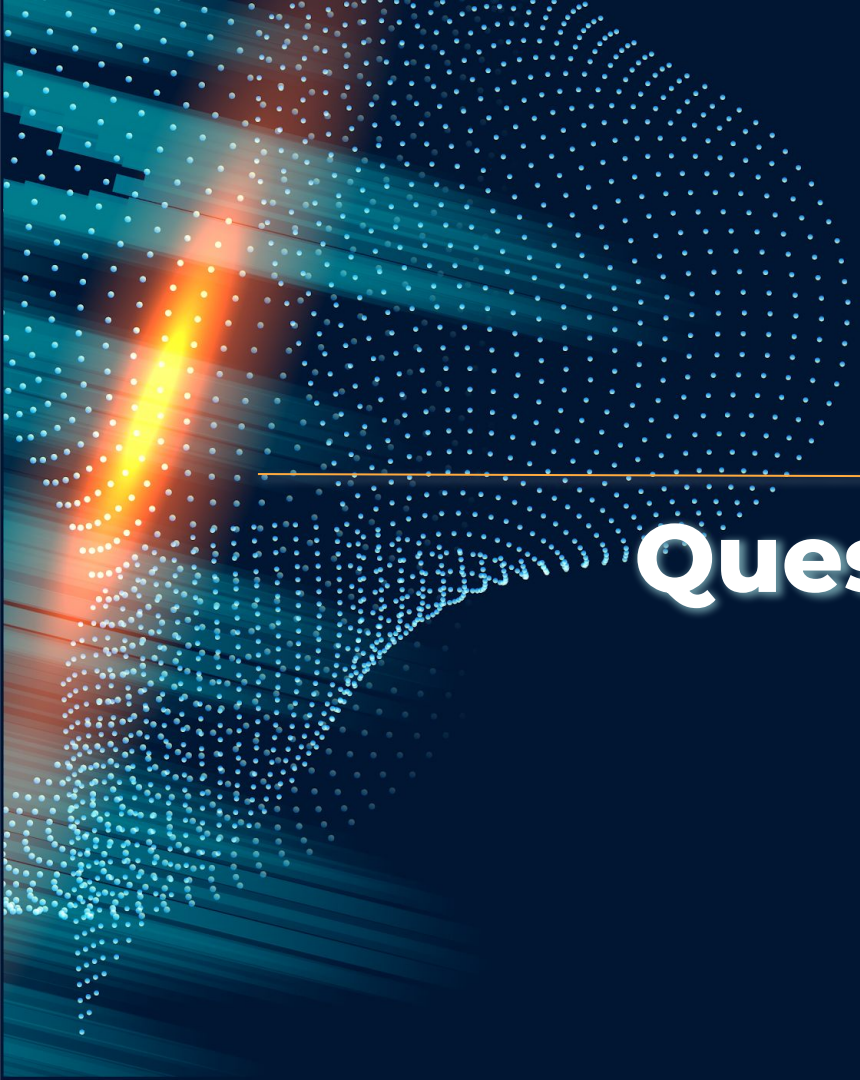
Timeline

29 september: abstract + presentation

1-20 october: additional group research

21 oct - 11 nov: implementation + testing

12 nov - 1 dec: write paper



Questions ?

Thank You

References

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- N Vanitha, G Padmavathi, “A Comparative Study on Communication Architecture of Unmanned Aerial Vehicles and Security Analysis of False Data Dissemination Attacks”, *Proceeding of 2018 IEEE International Conference on Current Trends toward Converging Technologies, Coimbatore, India.*
- Nesmah A. AL-Khulaidi, Ammar T. Zahary, Asma A. Al-shargabi, Muneer A.S Hazaa, “Machine Learning for Intrusion Detection in Vehicular Ad-hoc Networks (VANETs): A Survey”, *2024 4th International Conference on Emerging Smart Technologies and Applications (eSmarTA).*